Tractian Summer Internship 2026

TRACTIAN

GENERATIVE AI INTERNSHIP CASE

GENERATIVE AI INTERNSHIP CASE

CONTEXT

At Tractian, our goal is to make the day-to-day lives of on-site technicians and maintenance teams easier. Technicians can sometimes spend unnecessary effort digging through unstructured logs of previous machine issues, as well as dense physical and PDF OEM (Original Equipment Manufacturer) manuals.

Your objective is to design a **Retrieval-Augmented Generation (RAG) system** that instantly surfaces the relevant manufacturer guidelines and the historical context of past issues for a specific machine, along with helpful next step instructions.

Please outline the workflow, ingestion pipeline, and integration logic for this RAG-assisted feature **in either a 1-2 page document or a 3-5 slide presentation**. We will assess your submission based on structured thinking, relevance, practicality, and communication clarity. Your submission must detail the following points. A full implementation is not necessary but pseudocode is welcome if needed.

RAG INGEST & SEARCH STRATEGY:

Define how you will parse, chunk, and embed both the unstructured historical logs (written in markdown) of previous machine issues and the dense OEM manuals (saved in PDF files). Be sure to account for multimedia elements like images, charts, and graphs. Discuss your search retrieval strategy, including which models you would use for embedding and reranking and why.

VECTOR DB ARCHITECTURE:

Explain how you would organize the vector database. Your architecture must support two different entities: static OEM manuals and dynamic historical insights. Additionally, you must account for the fact that Tractian serves multiple distinct companies, and each company operates multiple different machines. Discuss your approach to scalability, data segregation, and embedding effectiveness (e.g., dimensionality, model architecture).

AGENTIC FRAMEWORK:

Lay out your conversational agent implementation. Explain why you chose your specific architecture or framework and defend why you selected it over other alternative options for this use case.

SESSION STORAGE:

Explain how your system will handle conversation history and state management during a back-and-forth troubleshooting session. Discuss your approach to session storage (e.g., memory window limits, database choice) to ensure the agent maintains context without exceeding token limits or losing critical diagnostic steps.

INPUTS & OUTPUTS:

Specify what input the LLM receives at inference time, and what structured output it produces to accurately assist the technician.

INTEGRATION:

Explain how this tool integrates into a workflow to reduce friction in the technician's day. Discuss the user interaction points and the appropriate notification cadence so that the tool is genuinely helpful without being intrusive.

RISKS:

Identify the risks or limitations of your design, and justify why accepting these risks is a reasonable trade-off given the benefits of your proposed system.

TIE TO TRACTIAN'S MOTION:

Tractian operates by turning real-world signals into structured decisions that drive action. In your submission, please explain why this specific RAG tool matters and what concrete decisions it supports for the maintenance team.